

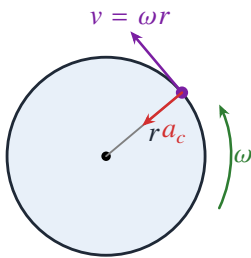
ROTATIONAL MOTION

Rotation Basics • Torque

JEE & NEET
PHYSICS

1 Angular Kinematics

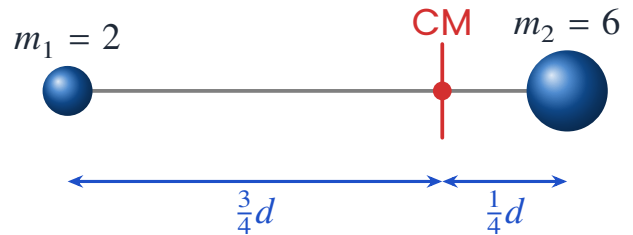
$$\omega = \omega_0 + \alpha t, \quad \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$
$$v = \omega r, \quad a_t = \alpha r, \quad a_c = \omega^2 r$$



Same equations as straight-line motion with $x \rightarrow \theta$.

2 Centre of Mass

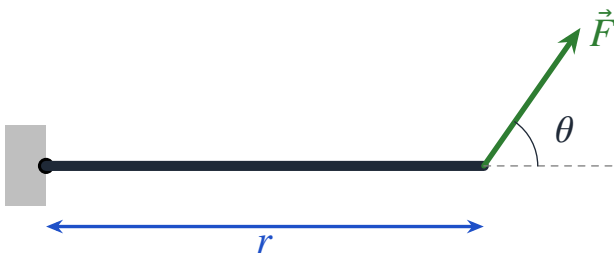
$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$



Closer to the heavier mass. Internal forces never move the CM.

3 Torque

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad \tau = rF \sin \theta$$



Only the part of F perpendicular to r turns the body.

4 Equilibrium

$$\Sigma \vec{F} = 0$$

$$\Sigma \vec{\tau} = 0 \text{ about any point}$$

A couple: zero net force but a turning effect.

ROTATIONAL MOTION

Moment of Inertia

JEE & NEET
PHYSICS

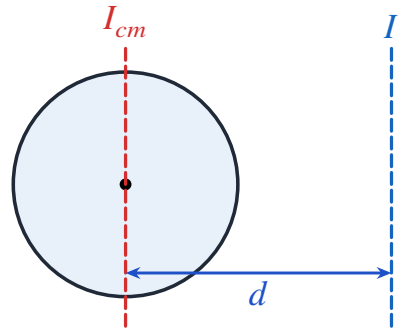
5 Moment of Inertia

$$I = \sum mr^2 = \int r^2 dm$$

Depends on the axis and on how far the mass is from it.

6 Parallel Axis Theorem

$$I = I_{cm} + Md^2$$



Rod about its end: $\frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{3}$.

7 Perpendicular Axis Theorem

$$I_z = I_x + I_y$$

Only for flat bodies. Ring about a diameter: $MR^2/2$.

8 Radius of Gyration

$$I = Mk^2, \quad k = \sqrt{I/M}$$

Disc: $k = R/\sqrt{2}$. Ring: $k = R$.

ROTATIONAL MOTION

Standard Bodies • Dynamics

JEE & NEET
PHYSICS

9 Moments of Inertia

Body (axis)	I	k^2/R^2
Ring (centre, perpendicular)	MR^2	1
Disc / solid cylinder (centre)	$\frac{1}{2}MR^2$	$\frac{1}{2}$
Hollow sphere (diameter)	$\frac{2}{3}MR^2$	$\frac{2}{3}$
Solid sphere (diameter)	$\frac{2}{5}MR^2$	$\frac{2}{5}$
Rod (centre / end)	$\frac{1}{12}ML^2 ; \frac{1}{3}ML^2$	--

10 Torque = $I\alpha$

$$\tau = I\alpha$$
$$a = \frac{mg}{m + M/2} \quad (\text{disc pulley})$$

Block m on a string over a disc pulley of mass M .

11 Rotational Energy and Power

$$K = \frac{1}{2}I\omega^2$$
$$W = \tau \theta, \quad P = \tau \omega$$

Same as $\frac{1}{2}mv^2$, Fs and Fv with rotation symbols.

ROTATIONAL MOTION

Angular Momentum • Rolling

JEE & NEET
PHYSICS

12 Angular Momentum

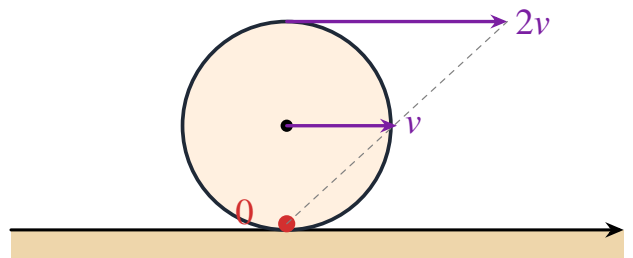
$$L = I\omega, \quad \vec{\tau} = \frac{d\vec{L}}{dt}$$

$$\tau = 0 \Rightarrow I_1\omega_1 = I_2\omega_2$$

A skater pulls her arms in: I falls, ω rises.

13 Pure Rolling

$$v_{cm} = \omega R$$



Top moves at $2v$, the contact point is at rest: friction does no work.

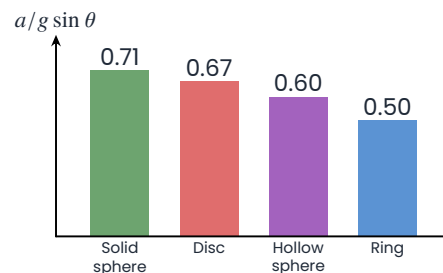
14 Kinetic Energy of Rolling

$$K = \frac{1}{2}Mv^2 \left(1 + \frac{k^2}{R^2} \right)$$

Rotational share = $\frac{k^2/R^2}{1+k^2/R^2}$: disc $\frac{1}{3}$, ring $\frac{1}{2}$.

15 Rolling Down an Incline

$$a = \frac{g \sin \theta}{1 + k^2/R^2}$$



Smaller k^2/R^2 wins; M and R do not matter.

ROTATIONAL MOTION

Quick Revision • Traps

JEE & NEET
PHYSICS

16 Linear vs Rotational

Linear	Rotational	Link
x, v, a	θ, ω, α	$v = \omega r$
m	$I = \Sigma mr^2$	$I = Mk^2$
$F = ma$	$\tau = I\alpha$	$\tau = rF \sin \theta$
$p = mv$	$L = I\omega$	$L = mvr_{\perp}$
$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$	rolling: both

17 Common Traps

- I changes with the axis: say which axis.
- Perpendicular axis theorem: flat bodies only.
- In pure rolling the contact point is at rest.
- The race winner does not depend on M or R .

18 Quick Numerical

A disc rolls without slipping. What fraction of its kinetic energy is rotational?

$$\frac{K_{\text{rot}}}{K} = \frac{1/2}{1 + 1/2}$$

$$\frac{1}{3}$$